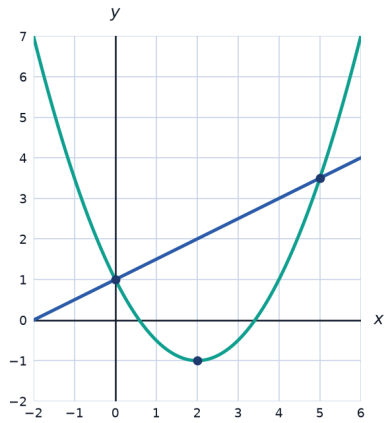
Mixed operations with integers

Core Concepts: Mixed operations with integers

1 For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(2)$.



a. 12

b. 14

c. 4

d. 16

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

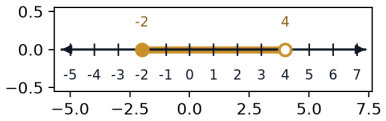
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(4)$.

a. 16

b. 18

c. 8

d. 20

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

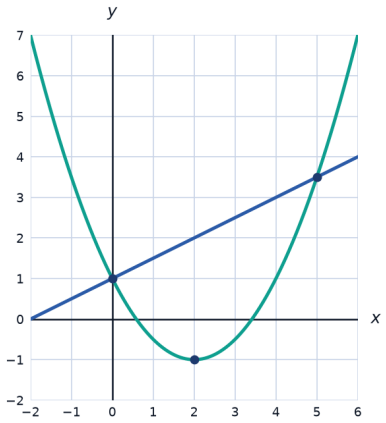
9 A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5



Exam-Style Problem Solving & Modeling

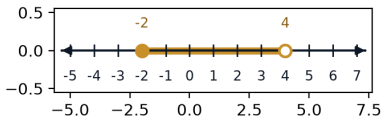
10 For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(5)$.

a. 18

b. 20

c. 10

d. 22



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647